

ICT 1209 – Web Technologies

Mini Project

# PlantIfi

An Interactive Botanical Database & Safety Guide

Rajarata University of Sri Lanka

Department of ICT

Course: ICT 1209 - Web Technologies

Batch: 2024 (First Year, Bachelor of ICT)

Assessment: Mini Project Phase 1 — Project Proposal & Wireframes

# PROJECT PROPOSAL

## 1. Project Title & Selected Theme

- **Project Title:** PlantIfi: An Interactive Botanical Database & Safety Guide
- **Selected Theme:** Innovative Theme — Botanical Directory (Details: Scientific names, toxicity profiling, edibility markers, ornamental plants and economic value)
- **Group Members:**
  - I.A.V.K. Ilangakoon (Index No: 2732)
  - B.S.C. Hasarangi (Index No: 2729)

## 2. Purpose and Goals

The idea behind PlanIfi comes from a very practical problem: biological data about plants is usually locked away in heavy textbooks or complex academic papers, while daily plant safety concerns affect almost everyone. Whether it is a home gardener trying to figure out if a backyard berry is edible, or a pet owner worrying about a houseplant their cat just chewed, quick and reliable answers are hard to find.

Our goal is to build an interactive, database-driven web application that organizes plants by their scientific profiles while highlighting everyday real-world details like toxicity, edibility, and commercial value. We want to make botanical knowledge accessible, help prevent accidental poisoning incidents, and educate users on the agricultural and economic potential of different plant species.

### 3. Target Audience

We designed this platform to serve a wide range of users, including:

- **Casual Gardeners and Foragers:** Everyday people who want to check if the plants in their living spaces or local environments are safe to handle, cultivate, or consume.
- **Agriculture & Students:** Peers who need a structured, searchable database of scientific names, classifications, and botanical traits for their own academic study and research.
- **Landscape Designers & Urban planners:** Professionals looking to verify safety profiles before introducing flora into public parks, schools, or pet-friendly spaces.
- **Eco-Tourists & Survivalists:** Outdoor enthusiasts who require a responsive field guide to distinguish edible species from dangerous look-alikes.
- **Agricultural Entrepreneurs:** Farmers and developers looking to evaluate high-yield crops based on their raw economic data, medicinal value, or manufacturing uses. (like timber and fibre).
- **Pet Owners & livestock farmers:** Individuals who need an instant, reliable index to verify if specific outdoor flora or indoor decorative plants are toxic to animals.

### 4. List of Planned Features

To hit all the core assessment requirements for this course, our application will implement the following functional features:

1. **Dynamic Plant Directory & Filter (Client-side JavaScript):** To ensure a smooth user experience, the main directory will use vanilla JavaScript to let user's sort and filter through the plant catalog (e.g., looking up only edible or toxic plants) instantly on the screen without needing to refresh the page.
2. **Featured Plant Interactive Slider (Client-side JavaScript):** The homepage will feature a responsive, interactive image showcase carousel that rotates through highlighted plants, focusing on critical safety warnings and high-value economic crops.
3. **Secure User Dashboard & Authentication (PHP & MySQL Backend):** We will build a complete login framework (login.php, register.php, logout.php) using secure

password hashing (PASSWORD\_BCRYPT). Once logged in, registered users can access deep-dive botanical profiles and save individual plants to a personalized bookmark dashboard.

4. **Validated Inquiry & Plant Report Form (JavaScript & PHP Backend):** A dedicated communication page (contact.php) featuring robust front-end JavaScript validation. This allows users to report unknown toxic plants or ask botanical questions, processing and storing the messages safely inside our MySQL database tables.

## 5. Proposed Database Organization

To keep our search filters lightning fast and organized, the database will tag each plant under one of five primary categories:

Edible Flora: Vegetables, fruit trees, and edible wild herbs.

Toxic / Poisonous Plants: Any species containing dangerous properties that pose a health risk to humans or animals upon contact or ingestion.

Medicinal Species: Organic plants historically or modernly used in pharmaceutical applications and home remedies.

Commercial / Economic Crops: Industrial plants grown specifically for timber, manufacturing fibers, rubber, or high-scale agricultural trade.

Ornamental Varieties: Popular indoor house plants, garden flowers, and aesthetic landscaping greenery.

## 6. Technical Architecture Stack

Our development process will follow the mandated course syllabus strictly, utilizing a completely local stack environment:

Frontend Design: HTML5, CSS3, and Bootstrap 5 to ensure the site is fully responsive on laptops, tablets, and mobile screens.

Client Side: Vanilla JavaScript for all UI dynamic interactions, animations, and form validation.

Backend Core: PHP 8 for processing forms, managing sessions, and handling backend server logic.

Database Management: MySQL, set up and managed locally through the XAMPP software system.

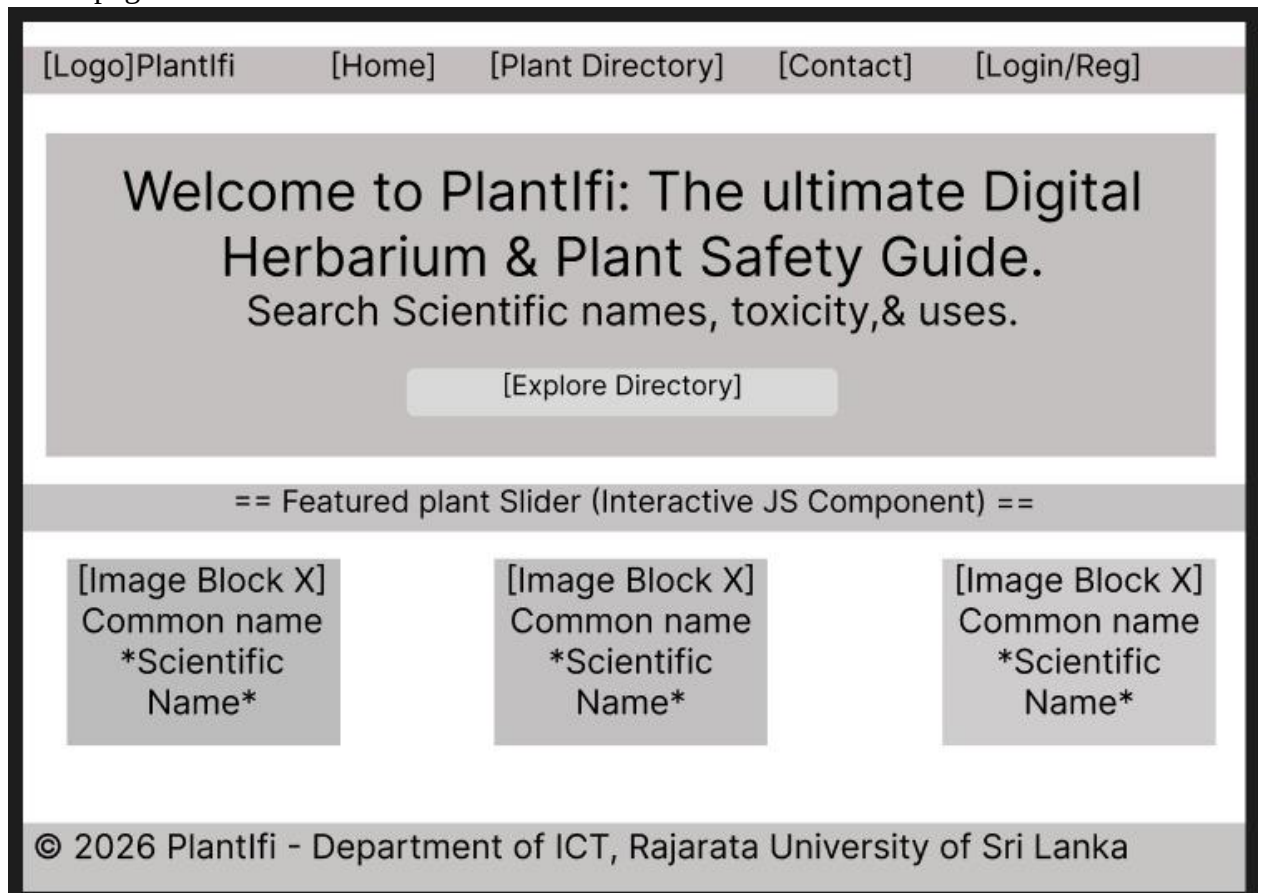
Version Control: Git and a shared public GitHub repository to track all our collaborative codebase additions and milestone commits.

## **7. GitHub Repository Link**

Project URL: <https://github.com/Basooriya/-Plantfi>

## 8. Wireframe

- Home page



- Plant Directive Grid & Dynamic Filters

[Logo]PlantIfi
[Home]
[Plant Directory]
[Contact]
[Login/Reg]

Search Plants: [ ] [Search]

Live Filters (JS):
☐ Edible
☐ Toxic
☐ Medical
☐ Economic

[Image Block X]  
Common name  
\*Scientific  
Name\*

[Image Block X]  
Common name  
\*Scientific  
Name\*

[Image Block X]  
Common name  
\*Scientific  
Name\*

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- Interactive Contact & Toxicity Report Form

[\[Logo\]](#) [Plantlfi](#)      [\[Home\]](#)    [\[Plant Directory\]](#)    [\[Contact\]](#)    [\[Login/Reg\]](#)

[\[Submit a Plant Toxicity Report/General Botanical Inquiry\]](#)

Your Name:   

Email Address:

\*(JS Real-Time Validation: Check for '@' format)

Inquiry Type:   

Plant Details:   

[\[ Submit Interactive Report \]](#)

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